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Executive Summary

In this exercise I inspected packets for VPN activity. Being able to identify traffic activities is crucial for network security engineers to identify suspicious behavior that might indicate attacks. In these examples I was looking for evidence of VPN traffic by identifying specific network protocols that were used along with some other attributes. For most known VPN protocols there are ports associated with that kind of traffic, the combination of a known VPN protocol with its known port is highly indicative of VPN traffic. Other things that I looked for were evidence of high traffic volumes between two specific addresses, indicating tunneling, also that the traffic itself was encrypted further indicating VPN traffic. For file 1 this was very straightforward because there was an obvious VPN protocol that was used with a known port repeatedly, which matches nearly all the criteria. Unfortunately I did not have to time to properly finish the rest of the files and will only be submitting the first file.

Technical Summary

File 1

For file 1 I see three clear periods of VPN like traffic. Packets 1 through 12, 21 and 22, and 27 and 28. All the packets mentioned are using the protocol 2TPv3 L2TP which typically works with UDP port 1701, which is visible in the packet details pane. The packets in the

capture show L2TP control messages, which indicate a VPN session setup and management. Additionally, the traffic is consistently between two addresses suggesting that tunneling is going on, another sign of VPN traffic. Finally, there are no visible application-layer protocols like HTTP, DNS, or SMTP suggesting encrypted tunneling. This combination of characteristics highly suggests that the packets identified are using a VPN.

1 0.000000	10.0.0.3	10.0.0.1	L2TPv3	185 Control Message - SCCRP (ccid=0x00000000)
2 0.000206	10.0.0.1	10.0.0.3	L2TPv3	171 Control Message - SCCRP (ccid=0x50CAD032)
3 0.000424	10.0.0.3	10.0.0.1	L2TPv3	62 Control Message - SCCCN (ccid=0x7736153E)
4 0.000486	10.0.0.1	10.0.0.3	L2TPv3	60 Control Message - ZLB (ccid=0x50CAD032)
5 0.000507	10.0.0.3	10.0.0.1	L2TPv3	131 Control Message - ICQP (ccid=0x7736153E)
6 0.000524	10.0.0.1	10.0.0.3	L2TPv3	98 Control Message - ICQP (ccid=0x50CAD032)
7 0.001123	10.0.0.3	10.0.0.1	L2TPv3	102 Control Message - ICCN (ccid=0x7736153E)
8 0.001247	10.0.0.1	10.0.0.3	L2TPv3	60 Control Message - ZLB (ccid=0x50CAD032)
9 0.001361	10.0.0.1	10.0.0.3	L2TPv3	92 Control Message - SLI (ccid=0x50CAD032)
10 0.001662	10.0.0.3	10.0.0.1	L2TPv3	100 Control Message - SLI (ccid=0x7736153E)
11 0.001779	10.0.0.1	10.0.0.3	L2TPv3	60 Control Message - ZLB (ccid=0x50CAD032)
12 0.013544	10.0.0.3	10.0.0.1	L2TPv3	54 Control Message - ZLB (ccid=0x7736153E)
13 47.413357	aa:bb:cc:00:04:00	Broadcast	ARP	110 D[5:0x8A57922B]who has 192.168.45.5? Tell 192.168.45.4
14 47.413363	aa:bb:cc:00:05:00	aa:bb:cc:00:04:00	ARP	110 D[5:0xAF94BC04]192.168.45.5 is at aa:bb:cc:00:05:00
15 49.416960	192.168.45.4	192.168.45.5	ICMP	164 D[5:0x8A57922B]Echo (ping) request id=0x0018, seq=1/256, ttl=255 (reply in 16)
16 49.417564	192.168.45.5	192.168.45.4	ICMP	164 D[5:0xAF94BC04]Echo (ping) reply id=0x0018, seq=1/256, ttl=255 (request in 15)
17 56.883878	aa:bb:cc:00:04:00	aa:bb:cc:00:04:00	Broadcast	ARP
18 56.884319	aa:bb:cc:00:05:00	aa:bb:cc:00:04:00	ARP	114 D[5:0xAF94BC04]192.168.46.5 is at aa:bb:cc:00:05:00
19 58.885871	192.168.46.4	192.168.46.5	ICMP	168 D[5:0x8A57922B]Echo (ping) request id=0x0019, seq=1/256, ttl=255 (reply in 20)
20 58.885990	192.168.46.5	192.168.46.4	ICMP	168 D[5:0xAF94BC04]Echo (ping) reply id=0x0019, seq=1/256, ttl=255 (request in 19)
21 60.000542	10.0.0.1	10.0.0.3	L2TPv3	62 Control Message - Hello (ccid=0x50CAD032)
22 60.000964	10.0.0.3	10.0.0.1	L2TPv3	54 Control Message - ZLB (ccid=0x7736153E)
23 60.159742	aa:bb:cc:00:04:00	Broadcast	ARP	114 D[5:0x8A57922B]who has 192.168.47.5? Tell 192.168.47.4
24 60.160253	aa:bb:cc:00:05:00	aa:bb:cc:00:04:00	ARP	114 D[5:0xAF94BC04]192.168.47.5 is at aa:bb:cc:00:05:00
25 62.162925	192.168.47.4	192.168.47.5	ICMP	168 D[5:0x8A57922B]Echo (ping) request id=0x001a, seq=1/256, ttl=255 (reply in 26)
26 62.163740	192.168.47.5	192.168.47.4	ICMP	168 D[5:0xAF94BC04]Echo (ping) reply id=0x001a, seq=1/256, ttl=255 (request in 25)
27 69.369421	10.0.0.1	10.0.0.3	L2TPv3	102 Control Message - StopCCN (ccid=0x50CAD032)
28 69.370639	10.0.0.3	10.0.0.1	L2TPv3	54 Control Message - ZLB (ccid=0x7736153E)

> Frame 1: 185 bytes on wire (1480 bits), 185 bytes captured (1480 bits) on interface -, id 0

> Ethernet II, Src: aa:bb:cc:00:02:10 (aa:bb:cc:00:02:10), Dst: aa:bb:cc:00:01:10 (aa:bb:cc:00:01:10)

> Internet Protocol Version 4, Src: 10.0.0.3, Dst: 10.0.0.1

> User Datagram Protocol, Src Port: 1701, Dst Port: 1701

Source Port: 1701

Destination Port: 1701

Length: 151

Character: 0x0000 (non-printable)

0000 aa bb cc 00 01 10 aa bb cc 00 02 10 08 00 45 c0E-

0010 00 ab 05 12 00 00 fe 11 a2 6c 0a 00 00 03 0a 001.....

0020 00 01 06 a5 06 a5 00 97 00 00 c8 03 00 0f 00 00

0030 00 00 00 00 00 00 80 00 00 00 00 00 01 80 08X]..

0040 00 00 00 02 01 00 80 0a 00 00 00 03 00 00 00R3..

0050 00 0e 00 00 00 05 17 cd 92 11 de 58 5d 83 00 08Cisco System

0060 00 00 00 0c 11 38 80 08 00 00 07 52 33 00 19s, Inc.

0070 00 00 00 08 43 69 73 63 6f 20 53 79 73 74 65 6d

0080 73 2c 20 49 6e 63 2e 80 08 00 00 00 0a 04 00 80

File 2

Did not complete.

File 3

Did not complete.

File 4

Did not complete.